

Few-Shot Radar Person Identification via Tracking-Integrated InISAR and Pre-Trained Meta-Learning

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Abstract—Radar-based person identification (PID) offers a privacy-preserving alternative to vision systems, yet practical deployment remains difficult. Conventional angle-of-arrival processing produces sparse, noisy point clouds affected by motion defocusing and multipath artefacts, while existing learning methods require large labeled datasets, making adaptation to new individuals impractical. We address these challenges through a unified framework combining tracking-integrated Interferometric Inverse Synthetic Aperture Radar (InISAR) with pre-trained meta-learning. The InISAR pipeline reconstructs high-resolution 3D human point clouds while suppressing multipath ghosts, achieving 73.78% IoU and <5.5 cm height error on motion capture validation. For learning, we introduce PCTrans, a spatio-temporal transformer embedded in MAML with task-specific pre-trained initialization to enable rapid few-shot adaptation. On 10,371 sequences from 14 subjects, the system achieves 98.99% accuracy with sufficient data and improves few-shot cross-session accuracy from 32.05% to 70.0%, with pre-training alone contributing +15.6 pp. This demonstrates a practical few-shot radar PID.

Index Terms—Few-Shot Learning, Meta-Learning, Model-Agnostic Meta-Learning (MAML), Person Identification, Point Cloud Transformer, Radar Point Clouds, Gait Recognition

I. INTRODUCTION

PRVACY-PRESERVING remote person identification (PID) is increasingly important for surveillance and human-robot interaction. While video-based approaches [1] are widely used, they raise privacy concerns, face regulatory constraints, and degrade under low-light conditions, adverse weather, and occlusions. On the other hand, Millimeter-wave (mmWave) radar provides a privacy-preserving alternative by avoiding facial information while maintaining robustness to environmental factors [2], [3].

However, practical radar-based PID remains challenging. **First**, conventional angle-of-arrival processing produces sparse and noisy point clouds with limited angular resolution, motion-induced defocusing, and multipath ghost artifacts, particularly in indoor environments [4]. **Second**, existing learning-based methods [5], [6] require extensive labeled data (often >100 sequences per individual, corresponding to several hours of

recording), which limits scalability and makes adaptation to new subjects from few examples impractical.

We address these two challenges through a unified framework that combines tracking-integrated Interferometric ISAR (InISAR) with pre-trained meta-learning. Rather than treating sensing and learning independently, we jointly improve radar point cloud quality and few-shot generalization. Our contributions are threefold:

(1) **Radar domain:** We develop a tracking-integrated InISAR pipeline that reconstructs high-resolution 3D point clouds of articulated human motion while mitigating multipath artifacts in indoor environments, and validate its performance against motion capture data.

(2) **Machine learning domain:** We introduce a spatio-temporal transformer architecture (PCTrans) embedded in Model-Agnostic Meta-Learning (MAML) [7], and show that task-specific pre-trained initialization substantially improves few-shot adaptation for radar point cloud person identification.

(3) **System-level integration:** We demonstrate that combining high-quality 3D radar reconstruction with meta-learning enables practical few-shot radar-based PID, reducing the dependence on large subject-specific training datasets.

Section II reviews related work. Section III provides background. Sections IV and V detail our approach. Section VI presents results. Section VII concludes.

II. RELATED WORK

We briefly review prior work from two complementary perspectives: radar imaging methods that determine the quality and structure of reconstructed point clouds, and learning-based approaches that operate on such data for person identification.

3D Radar Imaging & Point Cloud Deep Learning. Early radar-based PID approaches relied on micro-Doppler spectrograms [8], which capture motion signatures but lack explicit 3D geometric structure. Interferometric ISAR (InISAR) techniques [9]–[11] have demonstrated motion compensation, CLEAN autofocus, and interferometry for large rigid targets such as ships and aircraft. However, these methods were developed for extended, rigid scatterers and have not been systematically validated on small, articulated human targets,

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where body-part motion, low-radar cross-section segments, and multipath reflections introduce severe reconstruction challenges. In particular, indoor environments generate ghost clusters and aspect-dependent sparsity that can significantly distort point cloud structure, limiting direct adoption of classical InISAR pipelines for human PID.

On the learning side, PointNet [12] introduced permutation-invariant processing of point clouds via shared MLPs and max-pooling, but does not explicitly model local geometric relationships. The Point Cloud Transformer (PCT) [13] improves spatial representation through neighbor embedding and offset-attention, enabling better capture of local structures. Nevertheless, most prior PCT-based studies focus on static object recognition [14], whereas radar PID requires handling temporally evolving, sparse, and noise-corrupted point clouds.

Radar PID & Meta-Learning. Recent radar PID systems [5], [6] combine PointNet-style encoders with temporal convolutions or Transformers to capture gait dynamics. Despite improved temporal modeling, these methods require large subject-specific datasets and show near-random performance on unseen individuals. Model-Agnostic Meta-Learning (MAML) [7] enables rapid task adaptation and has been explored for static point cloud tasks [15], [16]. Most existing studies initialize MAML randomly, whereas work in NLP [17] indicates that pre-trained initialization can significantly accelerate meta-learning. We extend this idea to radar point-cloud PID, combining task-specific pre-training with meta-learning to improve few-shot generalization.

III. PRELIMINARIES

A. Problem Formulation

Input: Sequence $\mathcal{S} = \{\mathcal{P}_t\}_{t=1}^T$ of T point clouds, where $\mathcal{P}_t = \{p_i\}_{i=1}^{N_t} \subset \mathbb{R}^5$ contains N_t points $p_i = (x_i, y_i, z_i, v_i, i_i)$ encoding 3D coordinates, Doppler velocity, and intensity. **Output:** Person identity $y \in \{1, \dots, C\}$. **Few-Shot:** N -way K -shot tasks with support set $\mathcal{D}^{\text{support}}$ (K sequences per N individuals) for adaptation and query set $\mathcal{D}^{\text{query}}$ for evaluation.

B. FMCW Radar and Point Cloud Challenges

FMCW radar measures range and Doppler via FFT along fast-time (within chirp) and slow-time (across chirps). ISAR synthesizes larger apertures using target motion to achieve cross-range resolution. Motion compensation removes phase offsets caused by articulated body motion to focus scattered energy. InISAR extends ISAR to 3D using phase differences between spatially separated antennas for elevation estimation.

Radar point clouds exhibit: (1) **Aspect-Dependent Density**—count varies with orientation and range attenuation ($\propto 1/r^4$), creating dense torso clusters and sparse limbs. (2) **Multipath Noise**—indoor reflections produce ghost clusters [4]. (3) **Near-Field Errors**—height estimation degrades within 4m due to phase unwrapping ambiguities. (4) **Data Scarcity**—methods require > 100 sequences per person.

C. Point Cloud Transformer and MAML

PCT [13] enhances Transformers with neighbor embedding via FPS and k -NN aggregate local context via:

$$\Delta F(p) = \text{concat}_{q \in \text{knn}(p, k)} (F(q) - F(p)) \quad (1)$$

Offset-Attention emphasizes local structure differences:

$$F_{\text{out}} = \text{OA}(F_{\text{in}}) = \text{LBR}(F_{\text{in}} - F_{\text{sa}}) + F_{\text{in}} \quad (2)$$

approximating discrete Laplacian $(I - A)F$. Temporal Transformers [18] model pairwise sequences via $\text{Attention}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_k})V$ with sinusoidal positional encodings, capturing long-range dependencies without gradient vanishing.

MAML [7] learns meta-parameters θ for rapid task adaptation via two loops: **Inner** adapts to task $\mathcal{T}_i$ via:

$$\theta'_i = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\mathcal{T}_i}(f_{\theta}; \mathcal{D}_i^{\text{support}}) \quad (3)$$

Outer updates θ based on query performance:

$$\theta \leftarrow \theta - \beta \nabla_{\theta} \sum_{i=1}^B \mathcal{L}_{\mathcal{T}_i}(f_{\theta'_i}; \mathcal{D}_i^{\text{query}}) \quad (4)$$

IV. 3D POINT CLOUD GENERATION VIA INISAR

Our InISAR pipeline reconstructs high-resolution 3D point clouds through six stages: (1) **Range Processing**—Fast-time FFT. (2) **ISAR Motion Compensation**—Image Contrast-Based Autofocusing (ICBA) estimates and removes motion-induced artifacts. (3) **Doppler processing**—Slow-time FFT. (4) **CFAR Segmentation**—Smallest-Of Cell-Averaging Constant False Alarm Rate (SOCA-CFAR) segments target regions. (5) **Interferometry**—Phase-based 3D reconstruction. (6) **Multi-Target Tracking**—Extended Kalman Filter (EKF) and joint probabilistic data association (JPDA) jointly mitigate multipath effects.

A. Range-Doppler Signal Model

FMCW radar transmits LFM chirps $f(\tau) = f_C + K\tau$ ($0 \leq \tau \leq T_C$), characterized by bandwidth B and slope $K = B/T_C$. Hanning windowing and 2D-FFT yield range resolution $\Delta r = c/(2B)$ and velocity resolution $\Delta v = \lambda_0/(2MT_{PRI})$, where T_{PRI} denotes pulse repetition interval. For each antenna element m , the received baseband signal is:

$$S_{R,m}(f, t) = \int_{\mathbf{r} \in \text{target}} \sigma(\mathbf{r}) \exp\left(-j \frac{4\pi f}{c} R_m(\mathbf{r}, t)\right) d\mathbf{r}, \quad (5)$$

where $\mathbf{r}$ is scatterer position, $\sigma(\mathbf{r})$ reflectivity, and R_m the range to antenna m .

B. Motion Compensation and Autofocusing

Targets motion acceleration during coherent processing interval smears scatter energy across cells, yielding defocused and low SNR signals. Following [10], [19], the radial distance is $R_m(\mathbf{r}, t) = R_0(t) + \mathbf{r} \cdot \mathbf{i}_{\text{LOS}}(t)$, where $R_0(t)$ is translational displacement and $\mathbf{i}_{\text{LOS}}(t)$ is the line-of-sight unit vector. We model translation:

$$R_0(t) = r_0 + \beta t + \gamma t^2 \quad (6)$$

and estimate motion parameters β, γ via ICBA maximizing image intensity contrast:

$$[\hat{\beta}, \hat{\gamma}] = \arg \max_{\beta, \gamma} \left\{ \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (I_i - \bar{I})^2}}{\bar{I}} \right\} \quad (7)$$

Range compression (fast-time FFT) yields the raw range profile $S_{R,m}(f, t)$. For antenna element m , the motion-compensated range profile is:

$$S_{C,m}(f, t) = S_{R,m}(f, t) \cdot \exp \left(j \frac{4\pi f}{c} (\hat{\beta}t + \hat{\gamma}t^2) \right) \quad (8)$$

These signals are then coherently combined across virtual antennas within each row using geometric phase corrections. A subsequent slow-time FFT on the combined signal produces focused ISAR images for the top and bottom rows.

C. CFAR Segmentation and Interferometry

SOCA-CFAR ($N_T = 6$, $P_{FA} = 10^{-10}$) segments targets. The top and bottom ISAR images are co-registered before interferometry. For the co-registered complex images I_1 (top) and I_2 (bottom), the interferometry yields phase difference ϕ :

$$I_1 \cdot I_2^* = |I_1| |I_2| \exp \left(j \frac{2\pi}{\lambda} \Delta R \right) \quad (9)$$

where $\Delta R = L \sin \theta$ is range difference and L is baseline. This gives:

$$\phi = \frac{2\pi}{\lambda} L \frac{h}{R} \quad h = \frac{\lambda \phi R}{2\pi L} \quad (10)$$

Using azimuth θ_{az} and elevation θ_{el} (obtained from horizontal and vertical interferometry), 3D coordinates are reconstructed:

$$x = R \cos \theta_{el} \sin \theta_{az} \quad y = R \cos \theta_{el} \cos \theta_{az} \quad z = R \sin \theta_{el} \quad (11)$$

Velocity components (v_x, v_y, v_z) are derived similarly.

D. Multi-Target Tracking

EKF with JPDA in the 3D spatial domain follows to filter multipath ghosts and maintain track continuity [11]. The state vector includes 3D position, velocity, and acceleration. Detections are gated via Mahalanobis distance with a chi-squared threshold. A maximum miss count of one frame prunes inconsistent tracks while handling temporary occlusions; increasing this count would tolerate longer occlusions but risks retaining more noise. Fig. 1 shows DBSCAN alone leaves multipath ghosts (a), while EKF-JPDA removes them (b).

E. InISAR Validation

We assess the InISAR pipeline using two indoor datasets. Height estimation on five subjects (164–185 cm) walking toward a 60 GHz radar ($B = 3.3$ GHz, $\Delta r = 4.55$ cm) shows errors < 5.5 cm (median 3.87 cm), approaching the range resolution limit. In contrast, prior InISAR work on large rigid targets reported height errors of 0.2–1.6 m [10], [11].

Spatial reconstruction, compared against MoCap ground truth, achieves 73.78% average IoU across five movement patterns (Table I, Fig. 2). Reduced scores for kicking (69.19%)

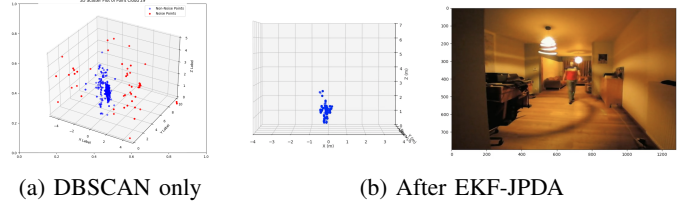


Fig. 1: Tracking comparison: (a) ghost targets remain; (b) ghosts removed.

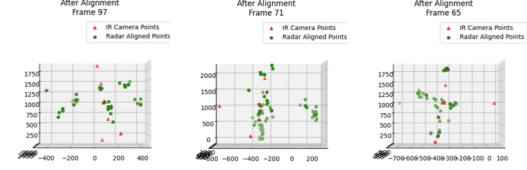


Fig. 2: InISAR vs. MoCap.

and squatting (59.98%) indicate current CFAR and tracking setting limitations in retaining transient scatterers during rapid acceleration change.

V. FEW-SHOT PERSON IDENTIFICATION

We address radar point cloud challenges through: (1) preprocessing, (2) PCTrans spatio-temporal feature extraction, and (3) pre-trained MAML.

A. Preprocessing

Preprocessing consists of four stages: (1) **DBSCAN** removes peripheral multipath through adaptive Kneedle algorithm ($k = 8$, $S = 1.0$). (2) **Near-field filtering** discards frames interfered by near-field phase ambiguity. (3) **Centroid normalization** shifts points to origin: $p'_i = p_i - (\bar{x}, \bar{y}, 0)$ to prevent overfitting to absolute location. (4) **Random padding** duplicates points to $n_{\max} = 1000$ [6]. After preprocessing, each sequence $\mathcal{S} = \{\mathcal{P}_t\}_{t=1}^{T=10}$ contains 10 frames with $\mathcal{P}_t \in \mathbb{R}^{1000 \times 5}$ having exactly 1000 centroid-normalized points with multipath and near-field artifacts removed.

B. PCTrans Architecture

PCTrans uses spatio-temporal feature extraction to reduce the impact of occluded body segments. Given preprocessed $\mathcal{S}$, we apply shared PCT encoder [13] frame-wise $\mathbf{f}_t = \text{PCT}(\mathcal{P}_t; \theta_{\text{PCT}})$, $t = 1, \dots, T$, where $\mathbf{f}_t \in \mathbb{R}^{192}$ is the spatial feature for frame t .

The PCT encoder applies two Sampling-and-Grouping (SG) layers with Farthest Point Sampling (FPS) and ball query: SG1 samples $n_1 = 200$ points with radius $r_1 = 0.2$; SG2 samples $n_2 = 100$ points with radius $r_2 = 0.3$. Ball query adapts to non-uniform density by grouping all points within a fixed spatial radius. Neighbor embedding and offset-attention then extract local geometric features. Frame-wise features $\{\mathbf{f}_t\}_{t=1}^T$ form a gait sequence. A Transformer encoder models temporal dependencies: positional encoding $\mathbf{f}_t^{\text{pos}} = \mathbf{f}_t + \text{PE}_t$ injects

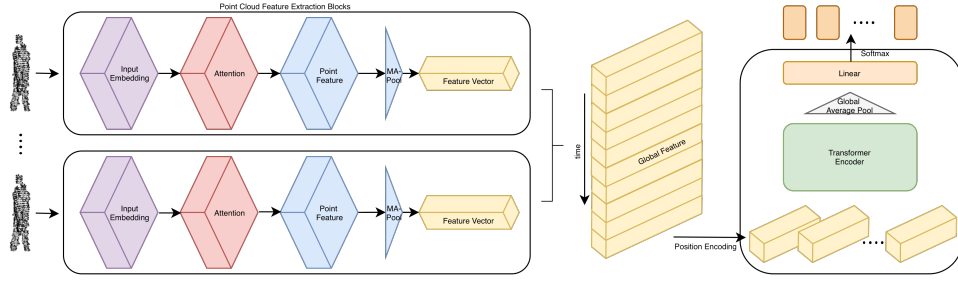


Fig. 3: PCTrans: Frame-wise PCT encoder extracts spatial features $\mathbf{f}_t \in \mathbb{R}^{192}$ and Temporal Transformer models gait dynamics

TABLE I: IoU Results for InISAR vs. MoCap

Capture Scenario	IoU (%)
Circular Walking (Cap 0)	79.82
Hands-Up Circular Walking (Cap 1)	100.0
Kicking (Cap 2)	69.19
Hands-Up + Kicking (Cap 3)	59.93
Squatting (Cap 4)	59.98
Average IoU	73.78

TABLE II: Supervised vs. MAML Performance

Query	TCPCN		PCTrans		MAML	
	S1	S3	S1	S3	S1	S3
Q1 (lin., same)	96.18	-	98.99	-	90.00	-
Q4 (non-lin., same)	-	48.03	-	51.16	-	47.78
Q2 (lin., cross-sess.)	33.03	35.10	32.05	31.67	56.67	43.44
Q3 (lin., cross-subj.)	51.16	57.81	40.09	31.99	51.11	36.67
Q5 (non-lin., cross-subj.)	31.42	30.54	31.10	29.46	37.78	27.78

temporal order, multi-head self-attention ($h = 4$ heads) captures cross-frame relationships, followed by max and average pooling to produces global feature vector $\mathbf{g} \in \mathbb{R}^{192}$. Finally, classification is performed via two FC layers with ReLU activation and Softmax:

$$p(y|S) = \text{Softmax}(W_2 \cdot \text{ReLU}(W_1 \mathbf{g} + b_1) + b_2) \quad (12)$$

C. Few-Shot Learning via MAML

We integrate PCTrans with MAML [7] for generalization. Each task $\mathcal{T}_i$ is 3-way 10-shot person identification: 3 subjects are classified using a support set with 10 sequences per subject and evaluated on a separate query set.

Meta-training follows [7]: **(1) Sample meta-batch**—Randomly sample $B = 3$ tasks from support set. **(2) Inner loop**—For each task, adapt θ via one gradient step (Eq. 3) with inner learning rate $\alpha = 0.1$ (SGD optimizer). **(3) Outer loop**—Update θ based on query performance (Eq. 4) with meta-learning rate $\beta = 0.001$ (Adam optimizer). The meta-training process is conducted for 1000 iterations, where learned θ enables rapid adaptation to new subjects.

D. Pre-Trained Meta-Parameter Initialization

Standard MAML initializes θ randomly (θ_0^{rand}). We propose pre-trained initialization, inspired by [17], **Two-Phase Algorithm**: Phase 1 trains PCTrans on representative task (e.g., 3 subjects) via supervised learning, saves θ_0^{PT} . Phase 2 initializes MAML with pre-trained weights, θ_0^{PT} , and is meta-trained across diverse tasks sampled from a broader dataset using Eqs. 3–4. This initialization reduces the number of meta-learning iterations required compared to random initialization.

VI. EXPERIMENTS

A. Dataset and Setup

We evaluate on both mmGait [20] and on 10,371 InISAR datasets from 14 subjects: linear (5,940) and non-linear (4,431)

motions, 10 frames/sequence, 64–1112 points/frame. Support sets $S1$ – $S4$ (3–14 subjects) train models; query sets $Q1$ – $Q5$ test same-session, cross-session, and (unseen-) cross-subject generalization. All experiments use PyTorch 1.12 with NVIDIA RTX 3090. Training: 10 hours; test-time: <1 second (6 gradient per iter).

B. Results

Benchmarking: On the public mmGait dataset [20], PCTrans achieves 95.21% vs. 78.33% (fixed vs. free) with 10 subjects, outperforming TCPCN [6] by +3.14pp vs. +8.02pp. With 20 subjects, PCTrans maintains 91.92% vs. 75.46%, surpassing TCPCN by +8.55pp (fixed) vs. +11.49pp (free). Meanwhile, when TCPCN is benchmarked against mmGaitNet on the mmGait dataset [20], TCPCN remains superior to mmGaitNet by +5.21pp (fixed) and +33.33pp (free). These findings confirm the architecture’s overall efficacy, but a direct comparison with our InISAR dataset is not comparable as mmGait’s “free” trajectories are less extreme (curved paths) than our high-acceleration actions (kicking, squatting), which our InISAR pipeline aggressively prunes to satisfy hardware limitations.

Supervised vs. MAML: Through supervised approaches, 90 epochs, and with SGD ($lr = 0.1$), PCTrans achieves 98.99% accuracy, outperforming TCPCN by 2.81% on Q1, but drops to near-random levels (31–33%) on cross-subjects and cross-session data (Table II). In contrast, Few-shot MAML-trained PCTrans, with minimal support sets ($S1$, $S3$: 3 subjects each), (1) sustained same-session performance at 90.00% on Q1 (vs. 98.99% supervised), and (2) improved cross-session generalization to 56.67% on Q2 (vs. 32.05% supervised).

Component Contribution: While a full ablation is beyond our scope, evidence suggests: (1) InISAR improves over conventional AoA [21] – mmGait fixed trajectory accuracy

TABLE III: Impact of Pre-Trained Initialization

Query	Random Init		Pre-Trained Init		Gain	
	S2	S4	S2	S4	S2	S4
Q1	76.67	-	94.44	-	+17.77	-
Q4	-	43.33	-	32.22	-	-11.11
Q2	54.44	37.79	70.00	42.22	+15.56	+4.43
Q3	46.67	38.87	48.89	47.78	+2.22	+8.91

(95.2%, no InISAR) vs. our InISAR data (98.99%). (2) Tracking/preprocessing remove ghosts and noise (Sec. IV). (3) Meta-learning alone yields an average +7pp (non-)linear cross-session and cross-subject gain (Table II).

Pre-Trained Initialization: MAML with θ_0^{rand} achieves 54.44%, while θ_0^{PT} enhances to **70.00%** (+15.56pp over random, +37.95pp over supervised), validating domain-aware θ_0^{PT} is closer to optimal θ^* than θ_0^{rand} . Gains are largest for cross-session (Q2: +15.56pp) as θ_0^{PT} encodes session-invariant patterns; cross-subject gains are modest (+2.22pp) as unseen gaits fall outside pre-training distribution (Table III).

PCTrans achieves 98.99% supervised (+2.81pp over TCPCN) and 70.0% few-shot cross-session (+38pp over supervised). Pre-training benefits generalization by encoding session-persistent gait patterns, enabling faster convergence and better few-shot adaptation. Due to intra-class variability, non-linear motions are less accurate than linear movements.

VII. CONCLUSION

We presented a comprehensive framework addressing radar-based person identification challenges: low angular resolution and data scarcity. Our tracking-integrated InISAR pipeline achieves 73.78% IoU and a less than 5.5 cm height error, approaching the 4.55cm range resolution limit, validated on articulated human targets in multipath-rich indoor environments. PCTrans with pre-trained MAML achieves 70.0% cross-session accuracy vs. 32.05% supervised baseline, representing a 38 percentage point (pp) improvement. Pre-trained initialization alone contributes +15.6 pp over random MAML. While our evaluation uses 14 subjects, future work will validate on larger, more diverse populations, explore larger meta-batch sizes, and develop adaptive tracking for high-acceleration scenarios. This framework makes radar-based person identification with minimal training data feasible.

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